

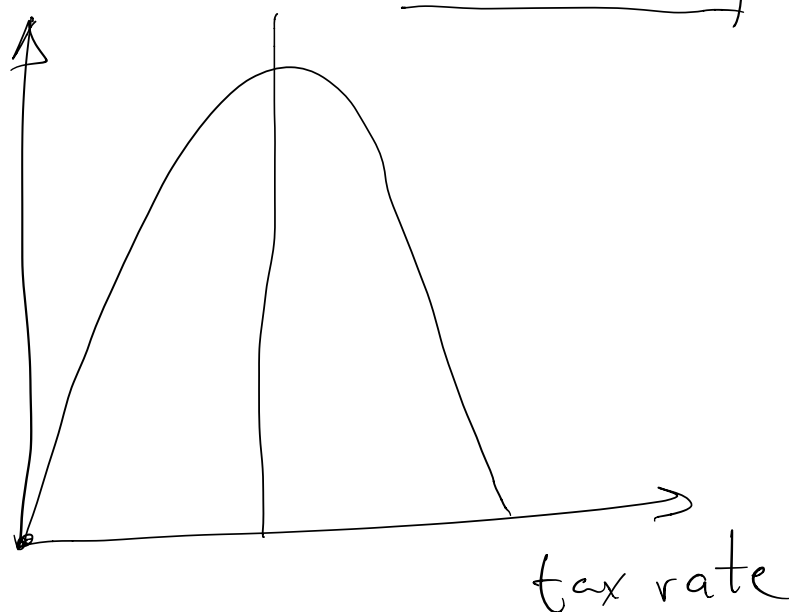
Notes Dyn Macro II
Topic 3.1
October 2020
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① Seigniorage is a Laffer Curve

Laffer:

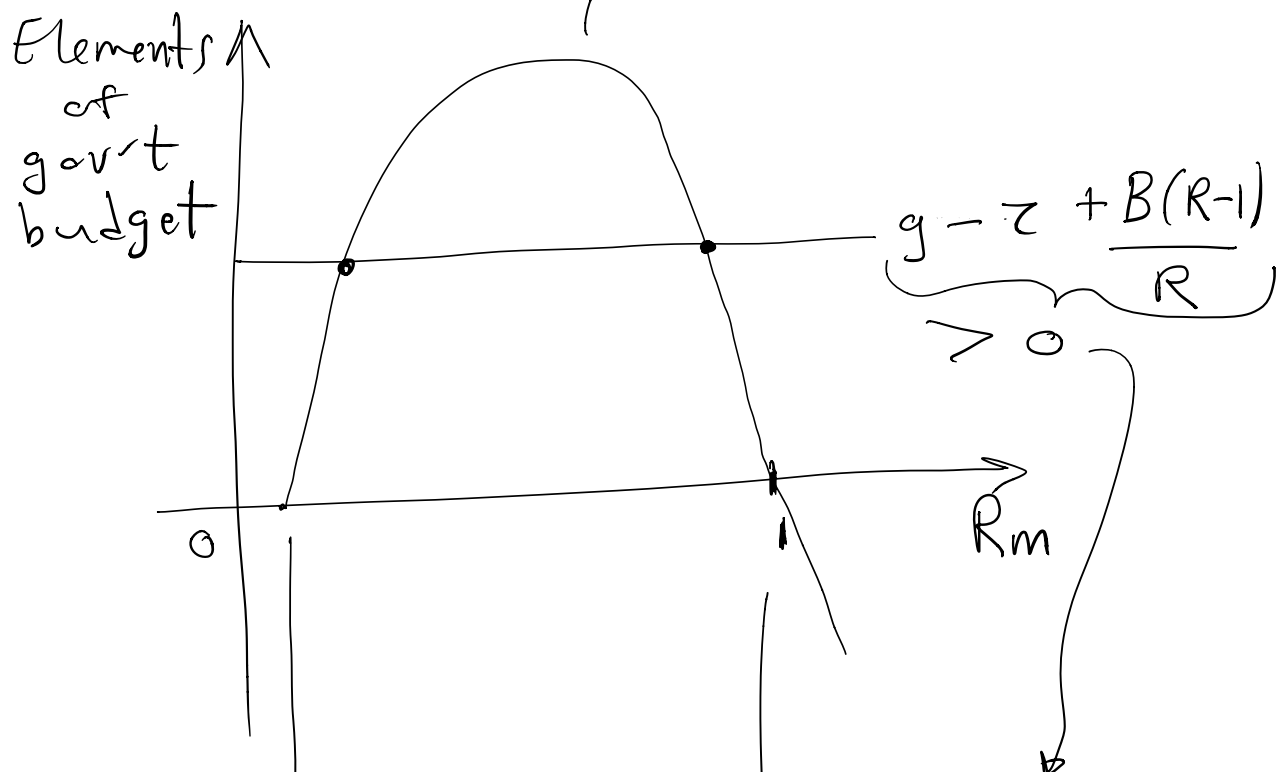
we can
construct
Laffer with
basic micro

tax
revenue:
a function
of τ



There exists a tax rate τ
that maximizes revenue

② Graphic solution of the model: Gov't budget for $t \geq 1$



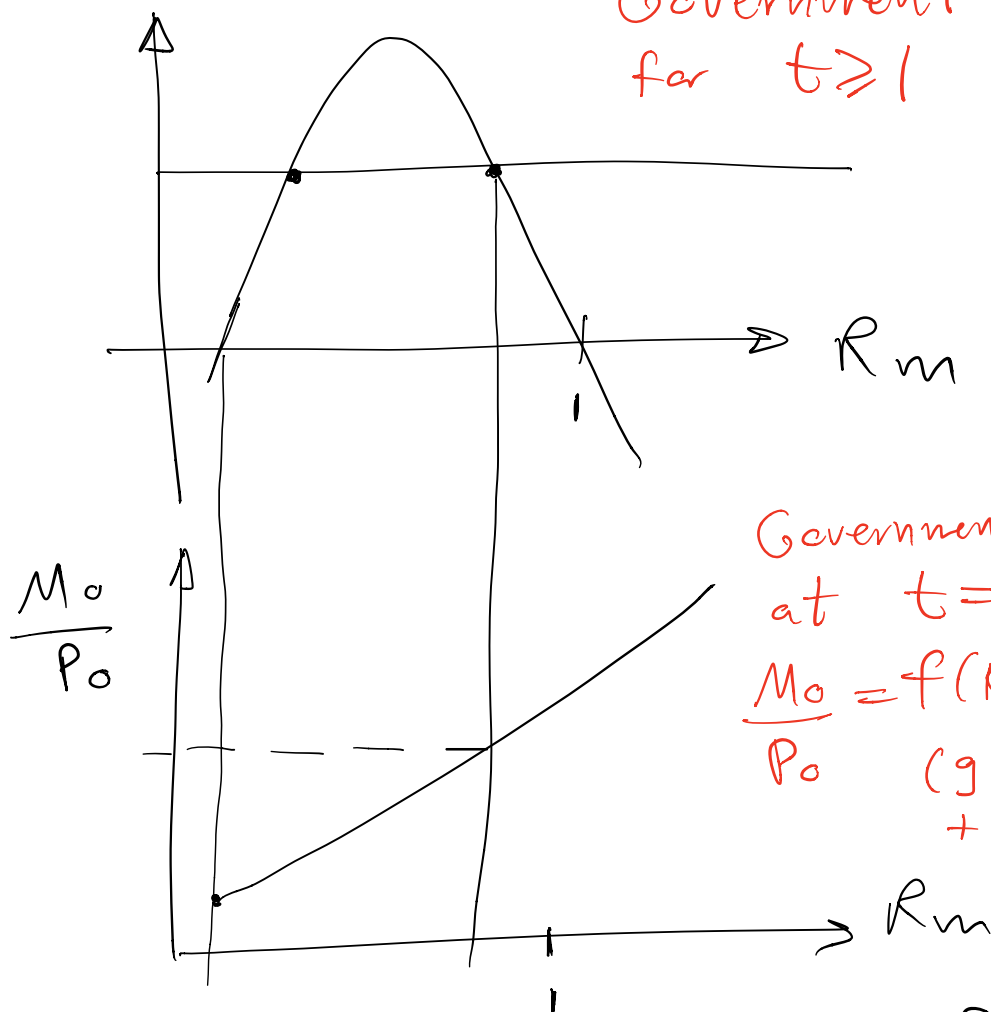
There may exist an R_m : $f(R_m) = 0$ (as in the example in the book)

I.e. there is a deficit (assumption)

With $R_m = 1$,
seigniorage = 0
with $R_m > 1$,
seigniorage < 0

Need to add a diagram below

Government budget
for $t \geq 1$



Government budget
at $t=0$:

$$\frac{M_0}{P_0} = f(R_m) - (g + B_0 - \tau_0) + B/R > 0$$

Take into account that $f'(R_m) \geq 0$

Example assumes $-(g + B_0 - \tau_0) + B/R > 0$

Otherwise M_0/P_0 could be < 0 ,
and we assumed $M_0 > 0$, $P_0 \geq 0$

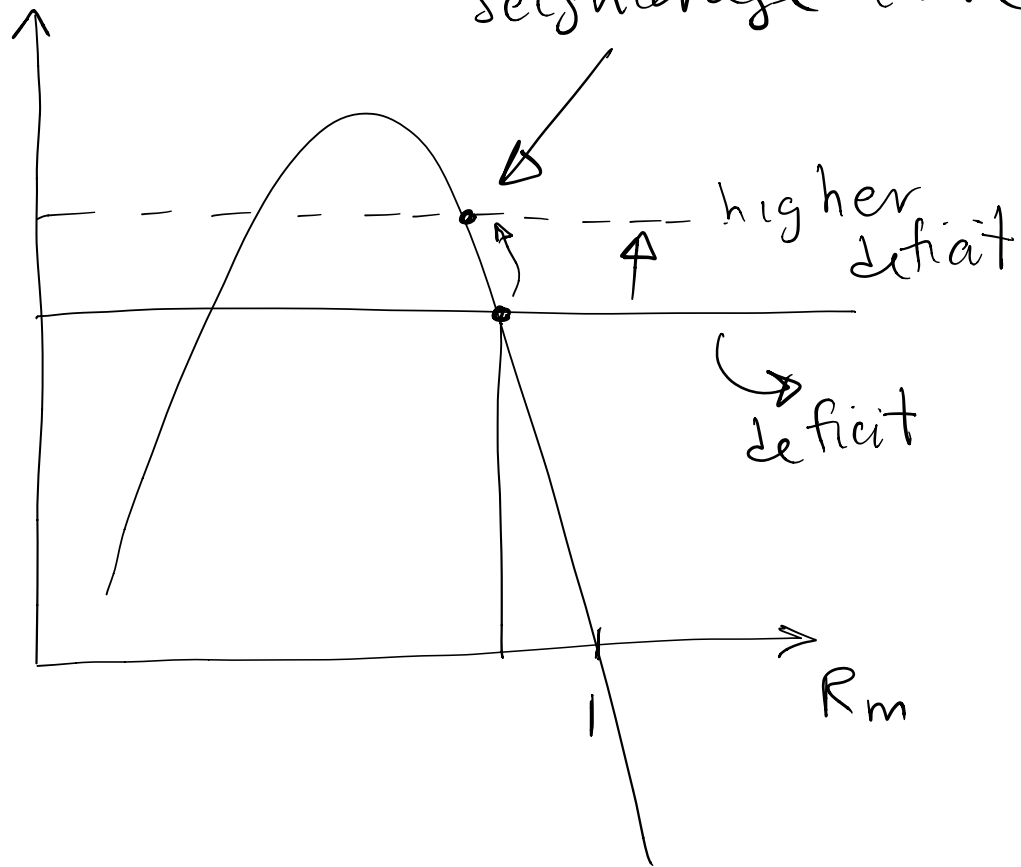
③ → In this simple model
there are 2 values for
seigniorage that satisfy the
budget
Which one do we choose?

The value to the right,
with a higher value for R_m ,
has a plausible dynamic:

an increase in the gross deficit
(including interest payments)
leads to a need for higher
seigniorage

Graphically:

Moving on the
Seigniorage curve

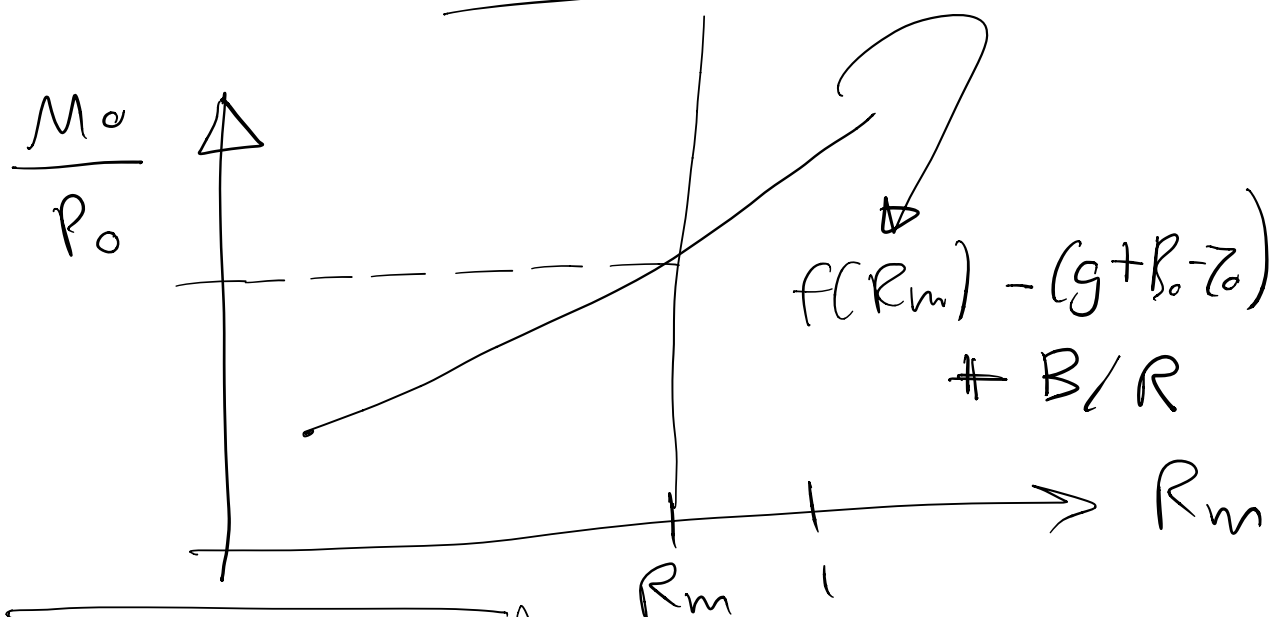


From now on we choose
the higher value of R_m
as the one that
satisfies the budget

Monetary "doctrines"

① Quantitative Theory
 M_0 does not determine

R_m
→ It does determine P_0

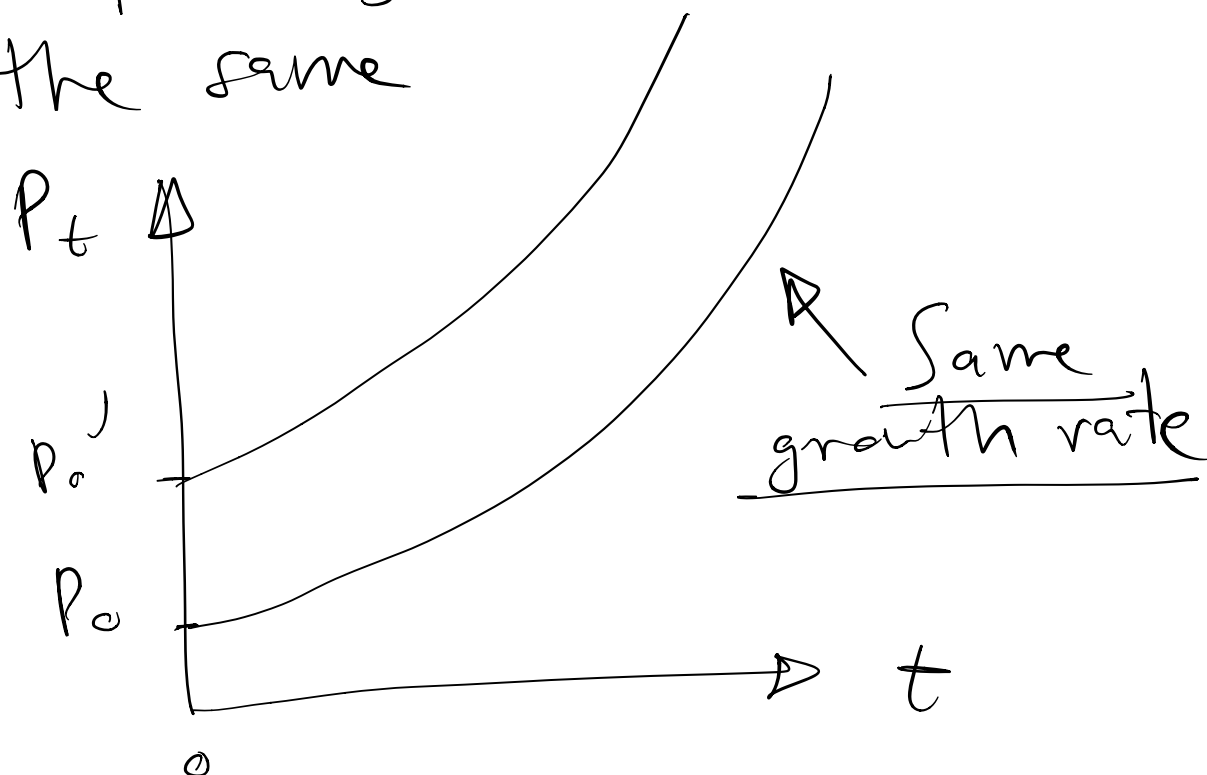


$\Delta M_0 \Rightarrow$
same ΔP_0 in
 P_0

↓
Determined in
previous graph

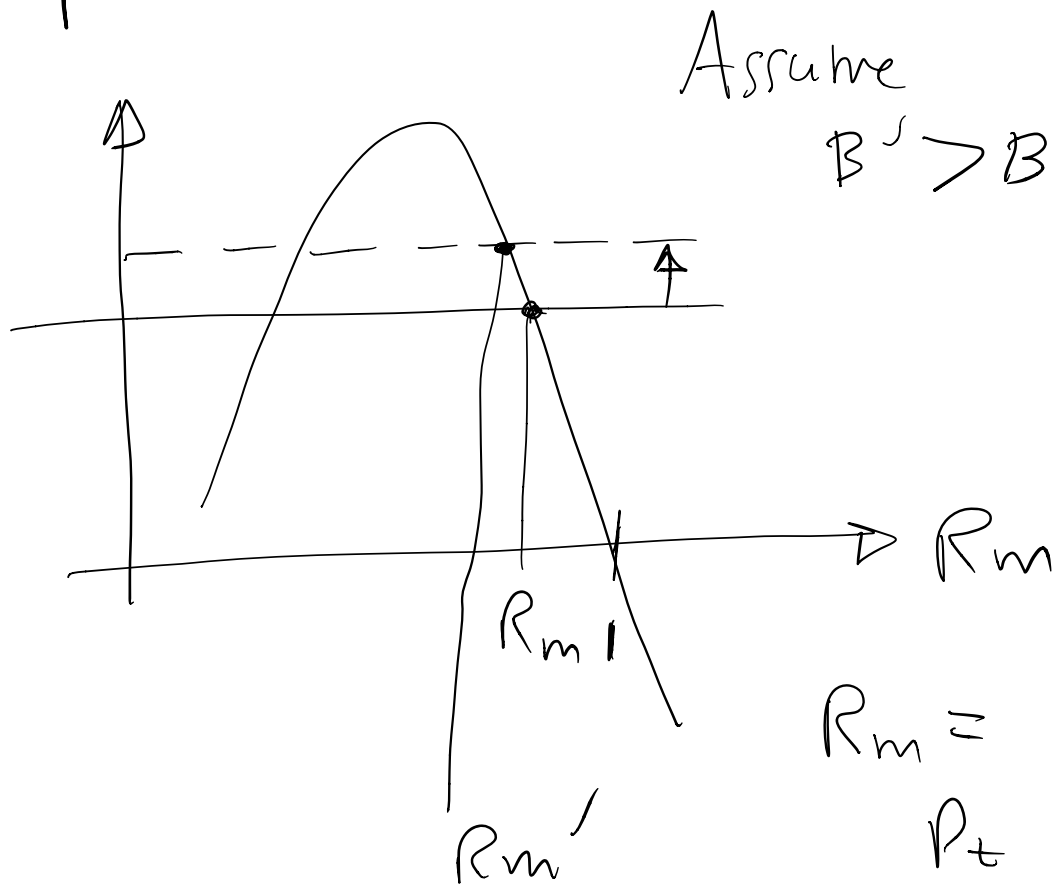
$\Rightarrow$ Price level is initially higher

The growth rate stays the same



② Deficits cause inflation

We have already seen the graph



$$R_m = \frac{P_t}{P_{t+1}}$$

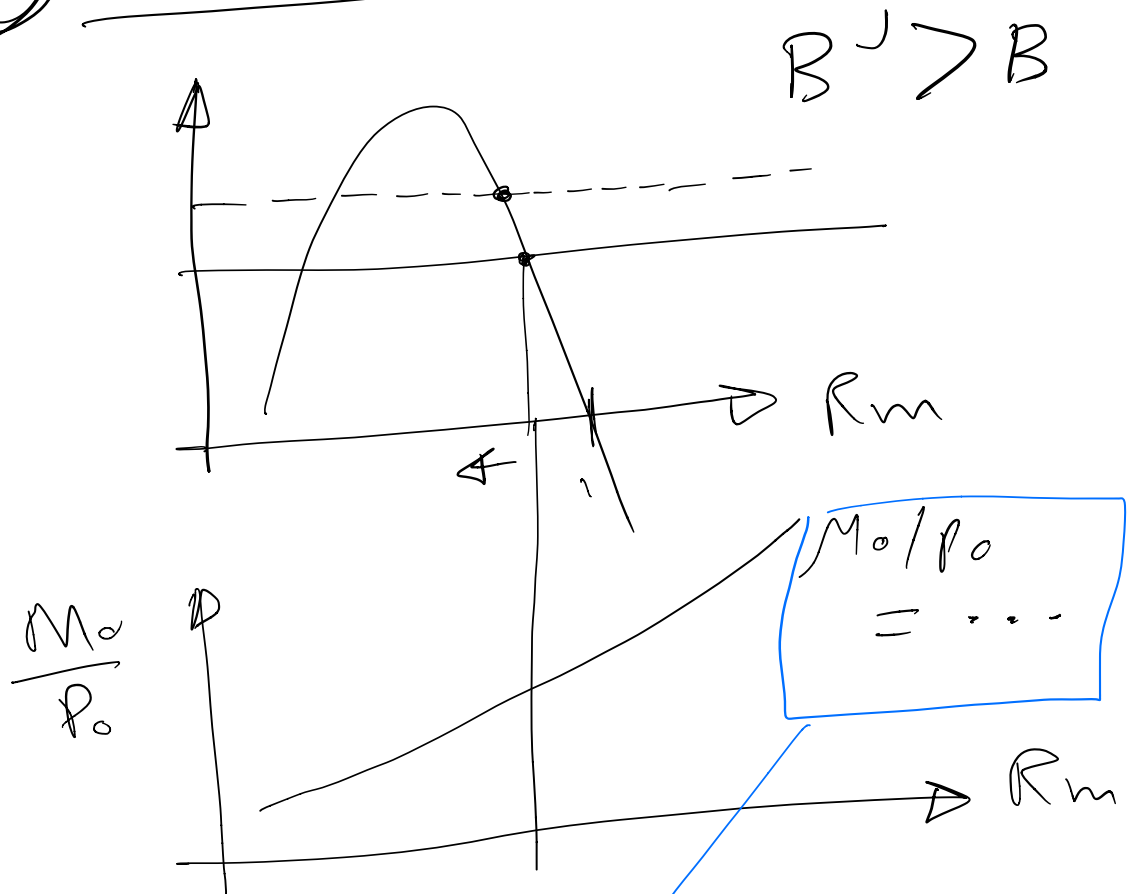
$$R_m' < R_m$$

$\Rightarrow$ inflation is higher

③ Fiscal prerequisites of zero inflation

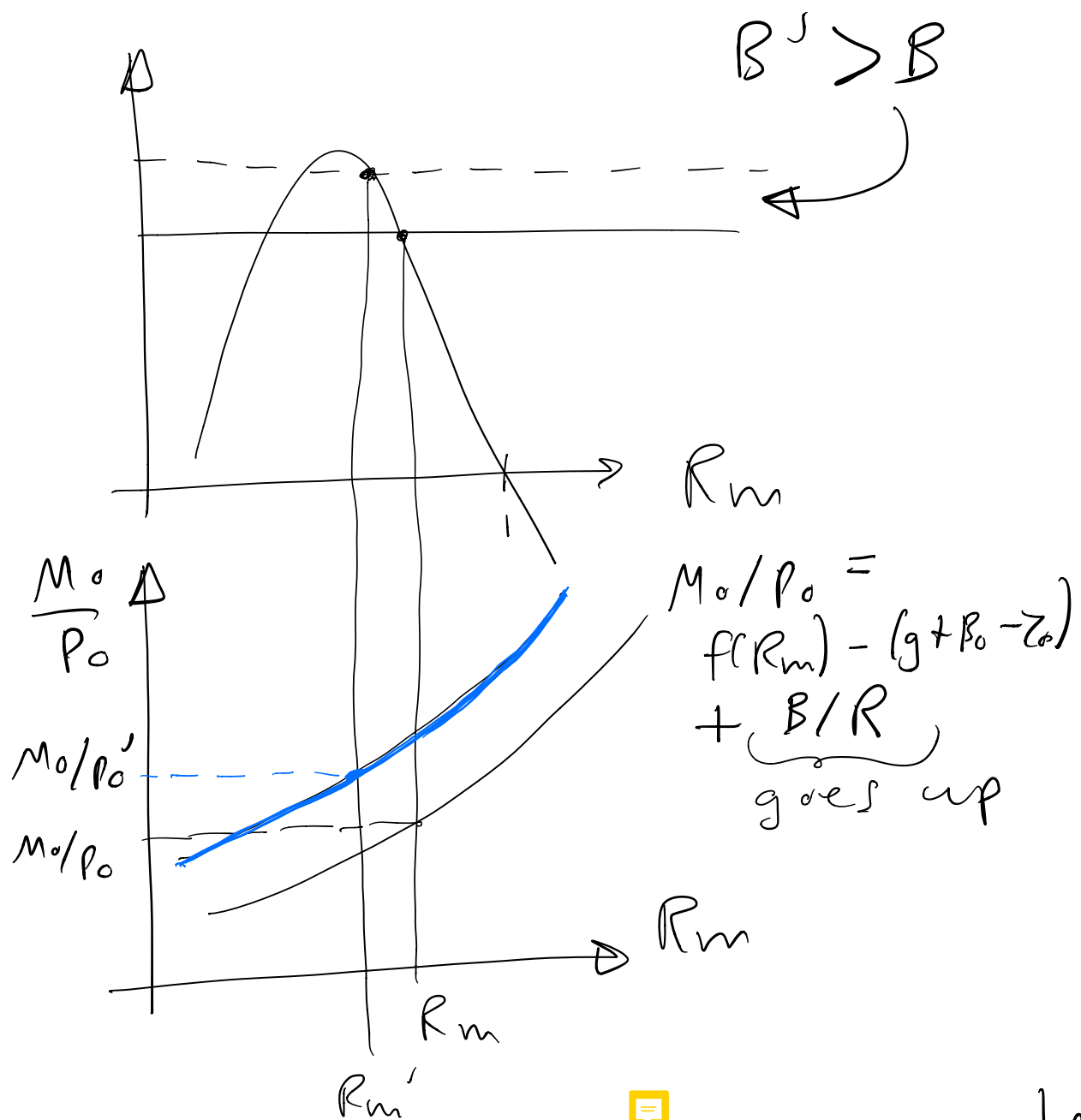
No graph

④ Unpleasant arithmetic



Depends on $B!$ $\Rightarrow$ it has to shift

Banko Central undi kemas $\Rightarrow \downarrow M_0 \uparrow B$



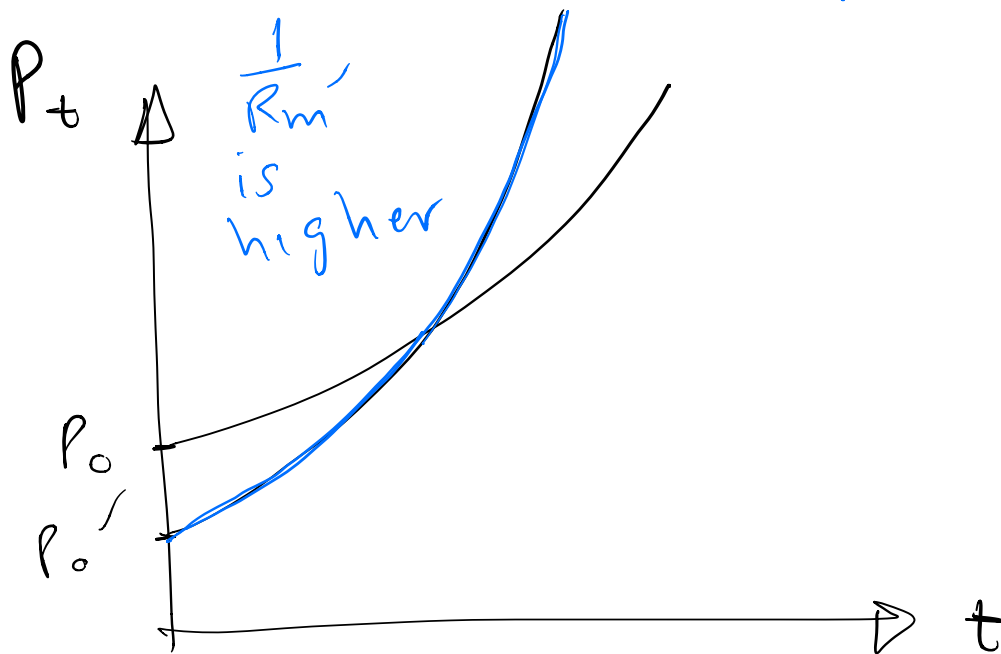
① $B' > B$ shifts curve upwards

② $R_m' > R_m$: movement on curve

In this example : $\frac{M_0}{P_0'} > \frac{M_0}{P_0} \Rightarrow \underline{\underline{P_0' < P_0}}$

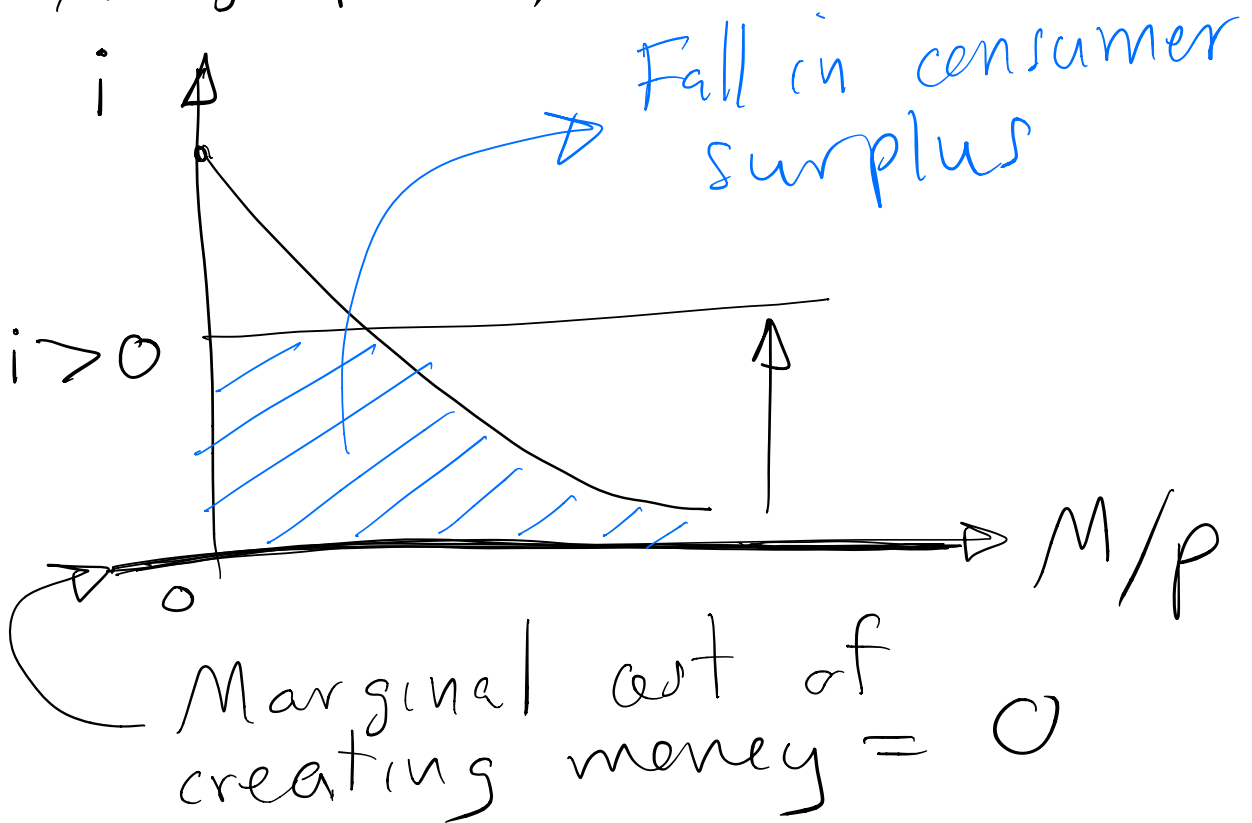
In this example,
price level is initially
lower

and "inflation" $= 1/R_m$
is higher (inflation
rate is $1/R_m - 1$)

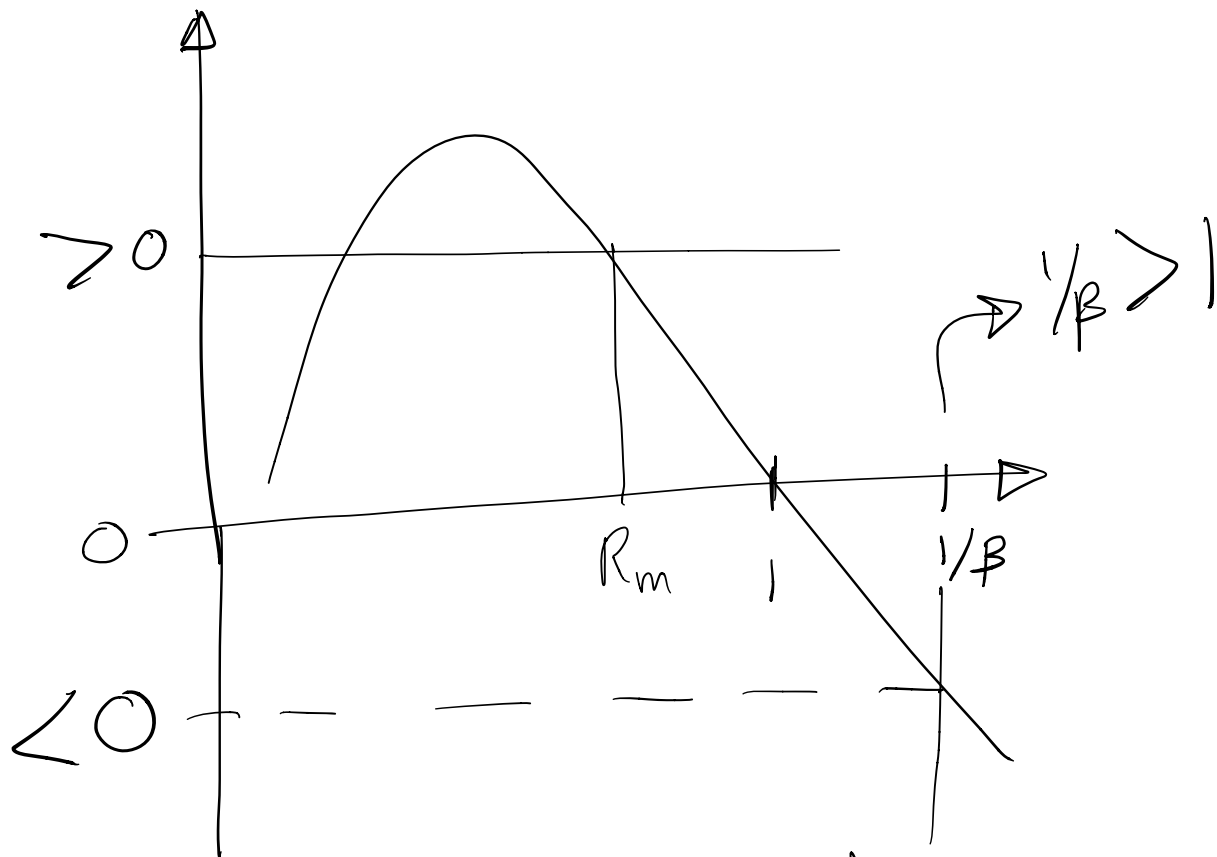


⑤ Optimum Quantity of Money

A graph by Friedman



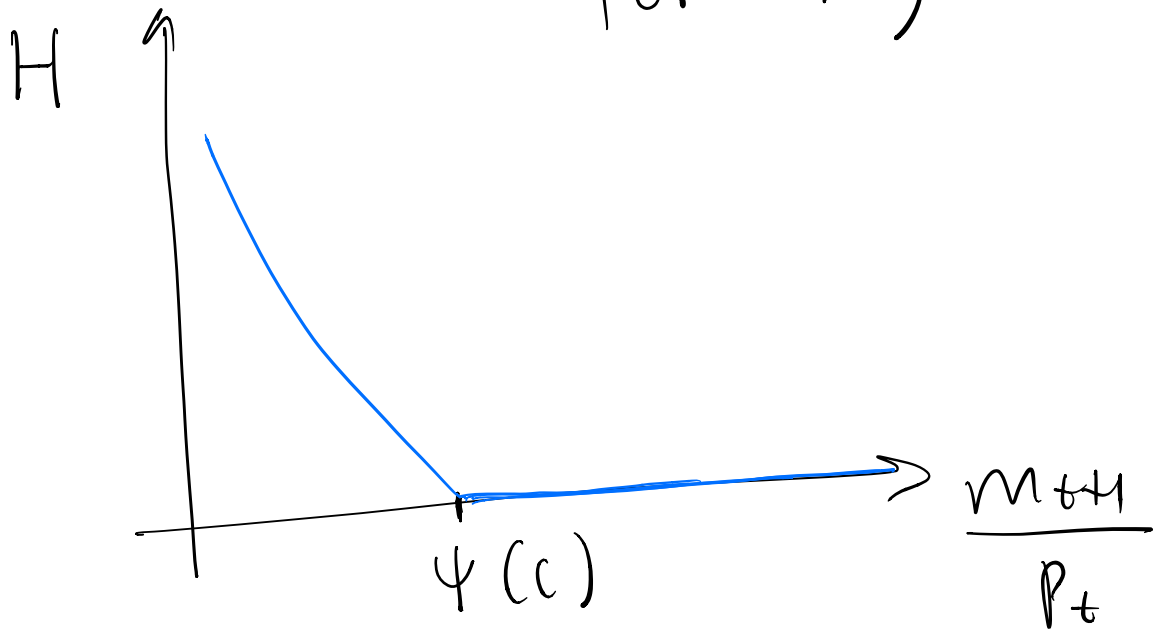
$$g - \tau + \frac{B(R-1)}{B} = f(R_m) \times (1-R_m)$$



To implement Friedman need
 $R_m \approx 1/B$
 Need $g - \tau + \frac{B(R-1)}{B} < 0$

I.e. need a suff^R large surplus

Graph for p. 14 in the slides (extra assumption for H)



(More generally,

